

Engineering Curves: Construction of Hypocycloid

Rolling Circle Method with Tangent & Normal

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Problem Statement

Problem

Draw a hypocycloid of a circle of diameter 50 mm, which rolls inside a circular arc of radius 85 mm for one revolution. Also, draw a tangent and a normal to the epicycloid at a point 50 mm from the centre of the directing circle.

Calculated Data

$$R = 85 \text{ mm},$$

$$OC_0 = R - r = 85 - 25 = 60 \text{ mm},$$

$$\angle AOB = 360^\circ \frac{r}{R} = 360^\circ \frac{25}{85} = 105.882353^\circ,$$

$$\Delta\theta = \frac{105.882353^\circ}{12} = 8.823529^\circ,$$

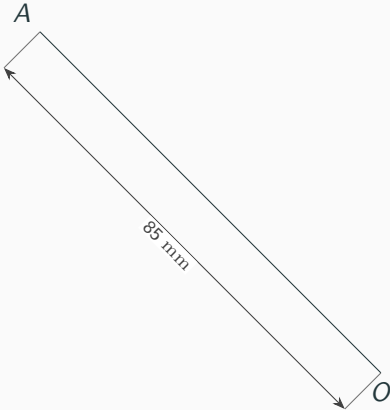
$$r = \frac{50}{2} = 25 \text{ mm},$$

$$AC_0 = 25 \text{ mm},$$

Generating-circle division = 30° .

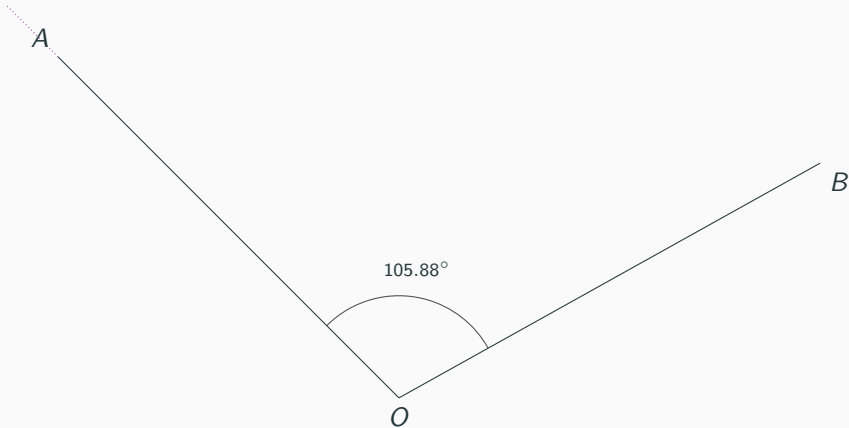
Step 1: Draw OA

$OA = 85 \text{ mm.}$



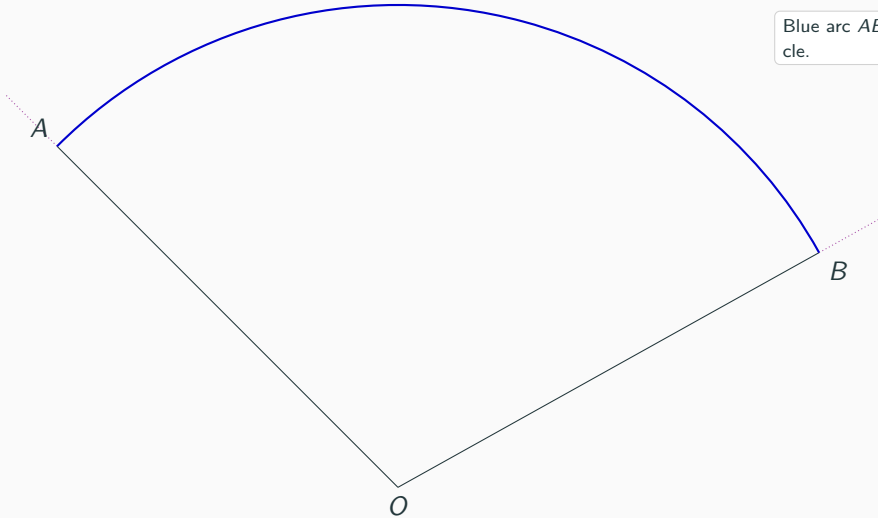
Step 2: Draw OB at 105.88°

$$\angle AOB = 105.88^\circ.$$



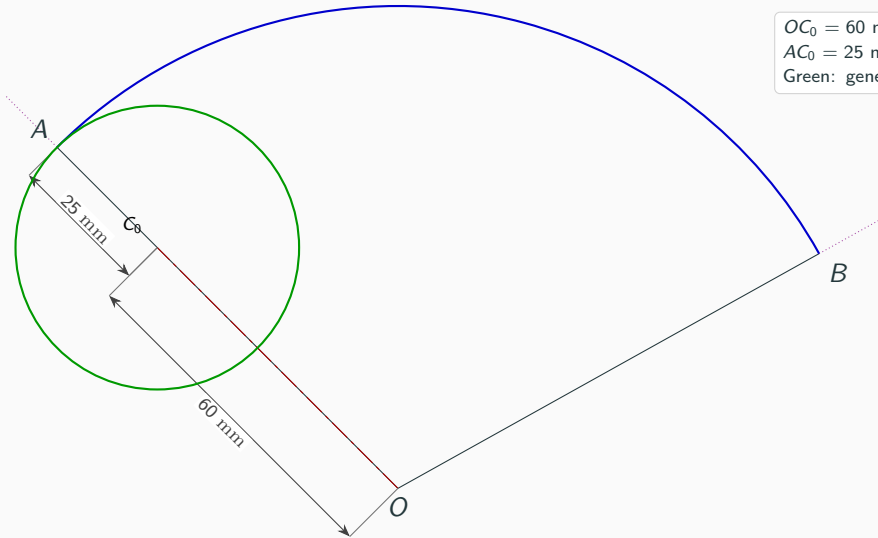
Step 3: Draw the directing arc AB

Blue arc AB : directing circle.



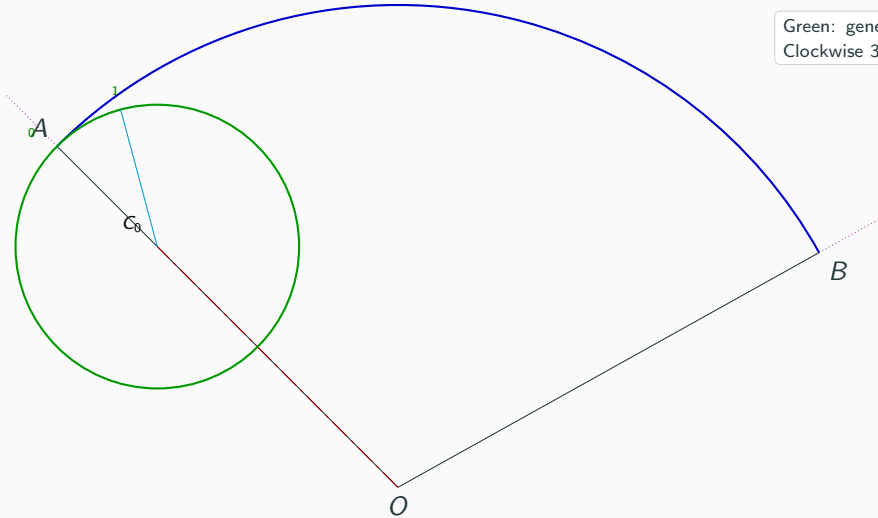
Step 4: Locate C_0 and the generating circle

$OC_0 = 60 \text{ mm.}$
 $AC_0 = 25 \text{ mm.}$
Green: generating circle.

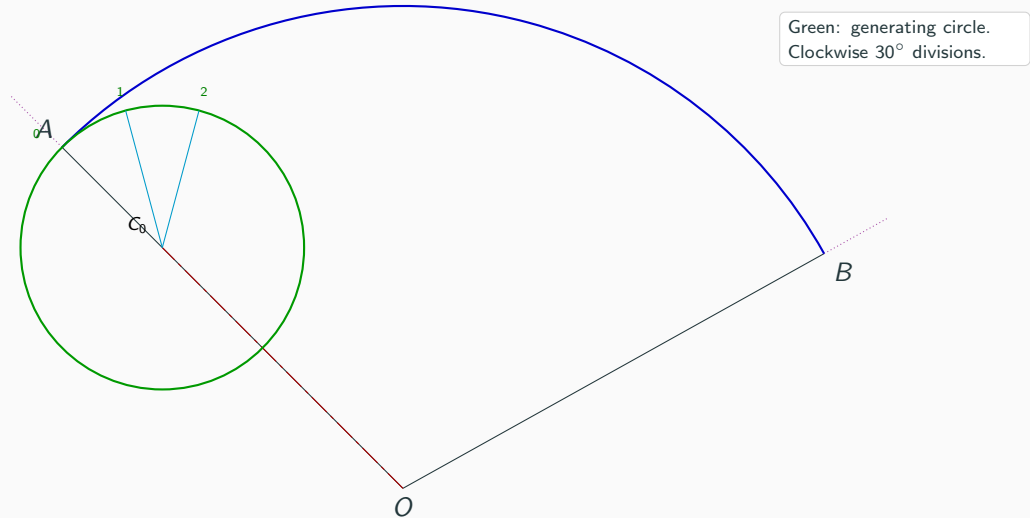


Step 5: Divide the generating circle into 12 equal parts

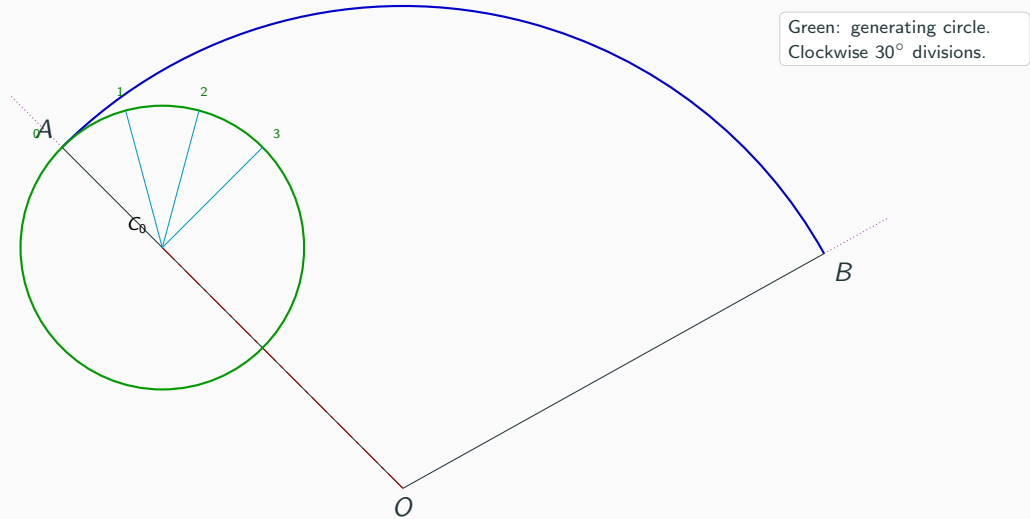
Green: generating circle.
Clockwise 30° divisions.



Step 5: Divide the generating circle into 12 equal parts

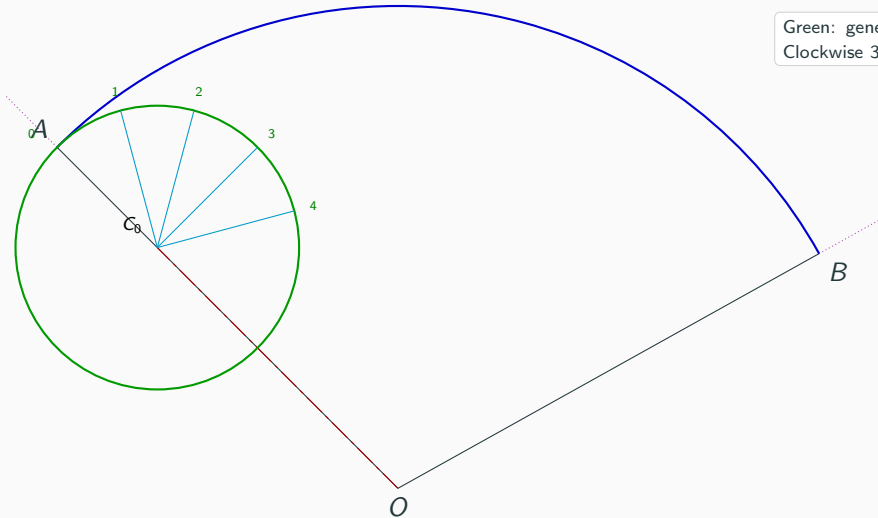


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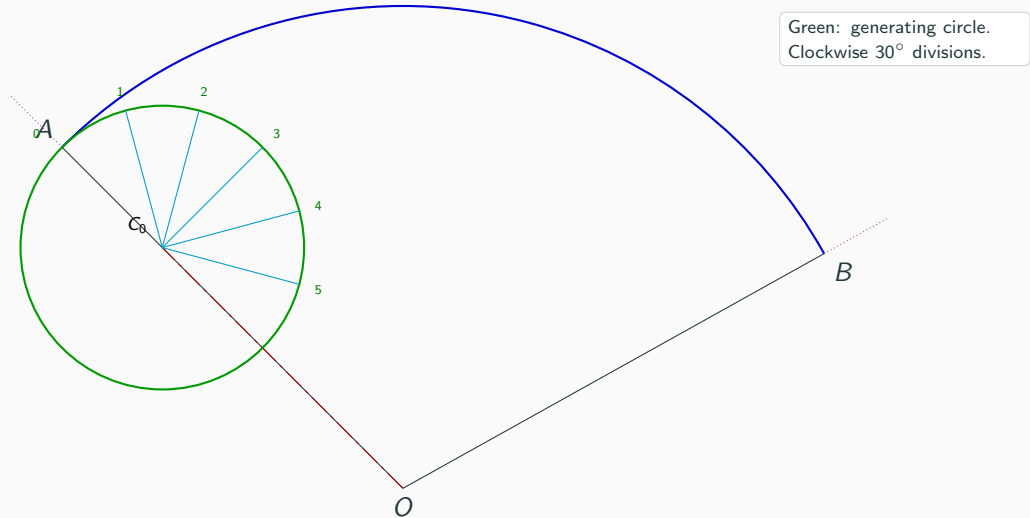


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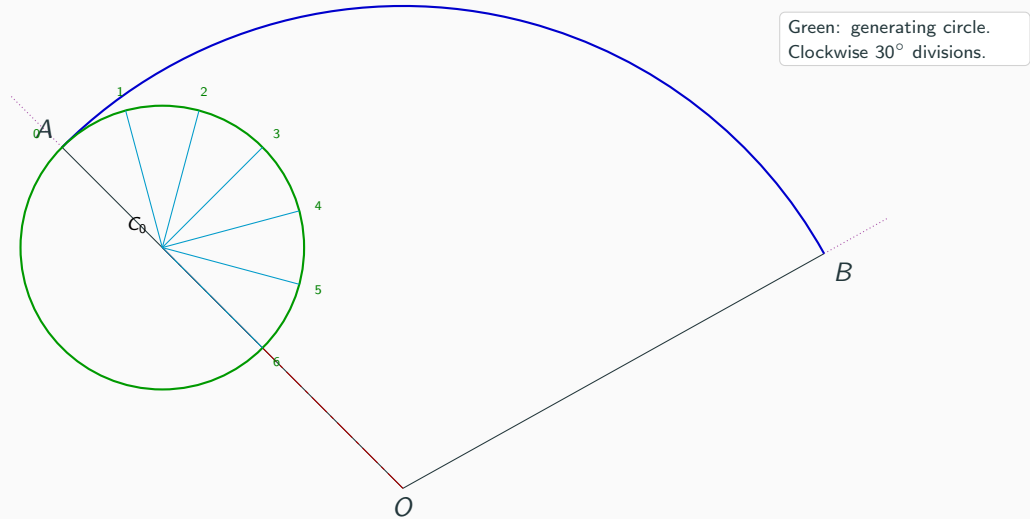
Green: generating circle.
Clockwise 30° divisions.



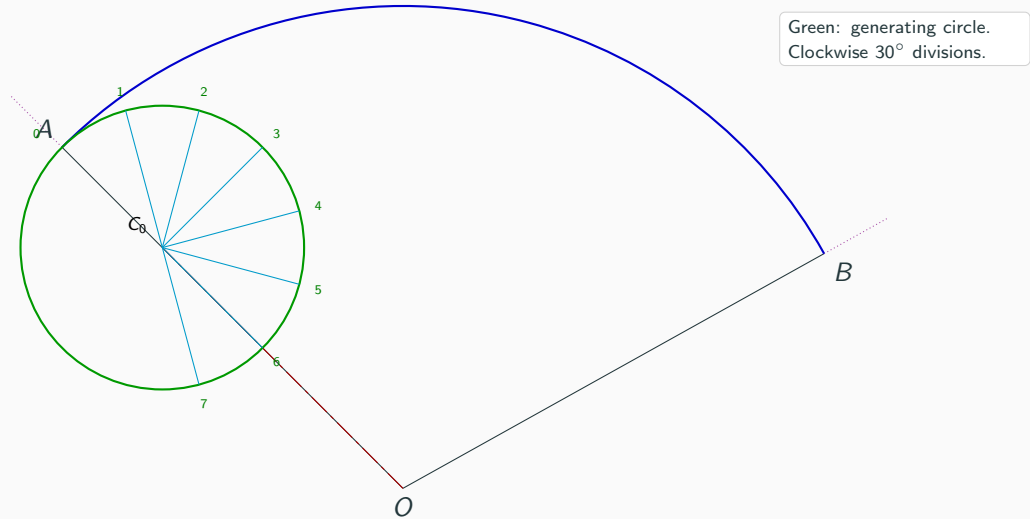
Step 5: Divide the generating circle into 12 equal parts



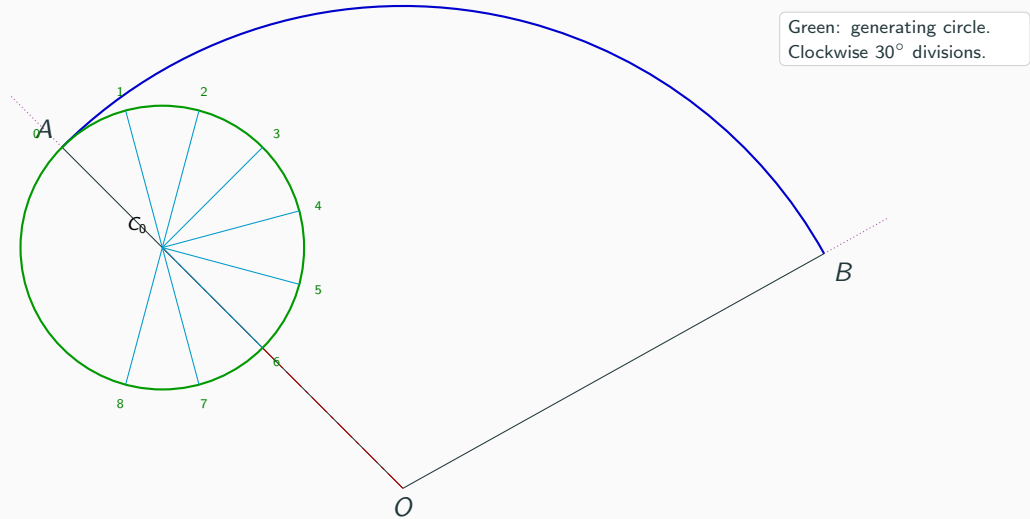
Step 5: Divide the generating circle into 12 equal parts



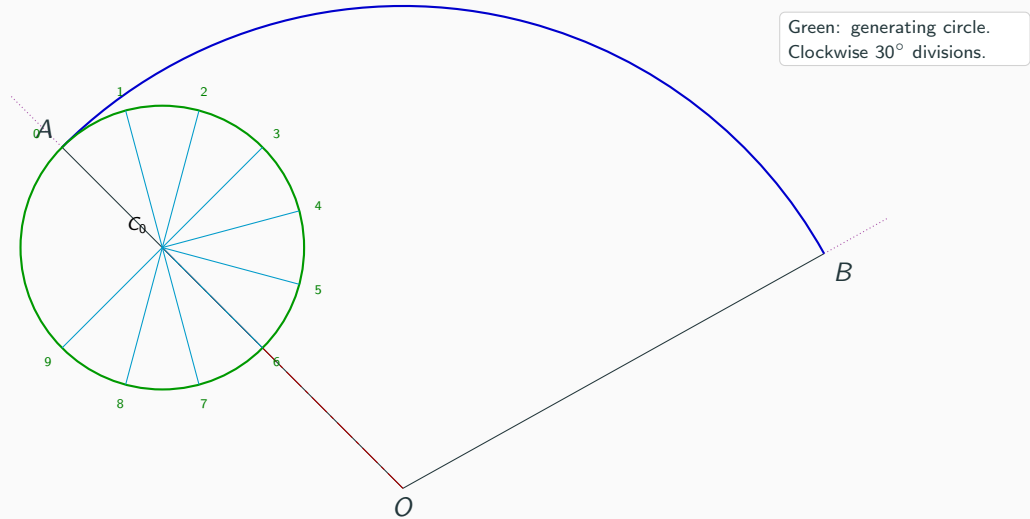
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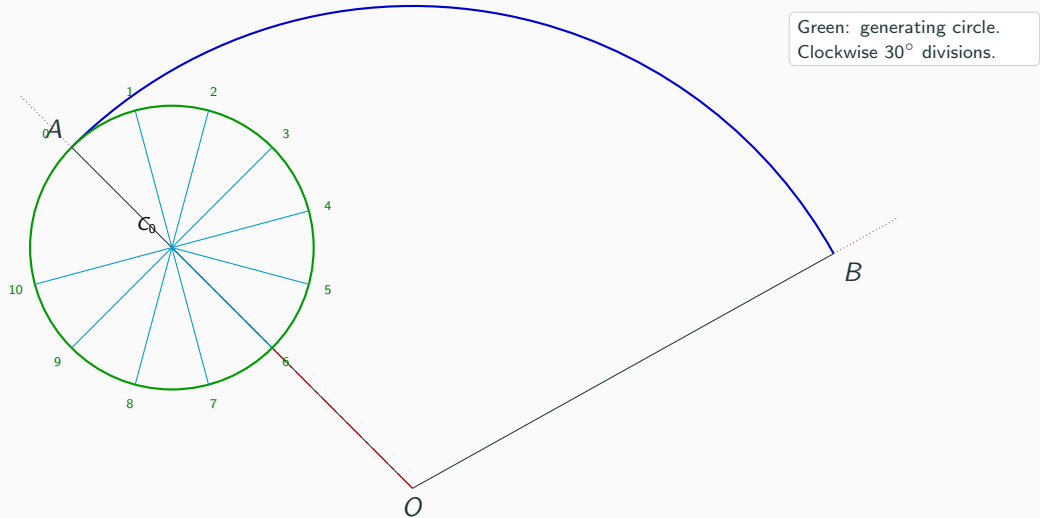
Step 5: Divide the generating circle into 12 equal parts



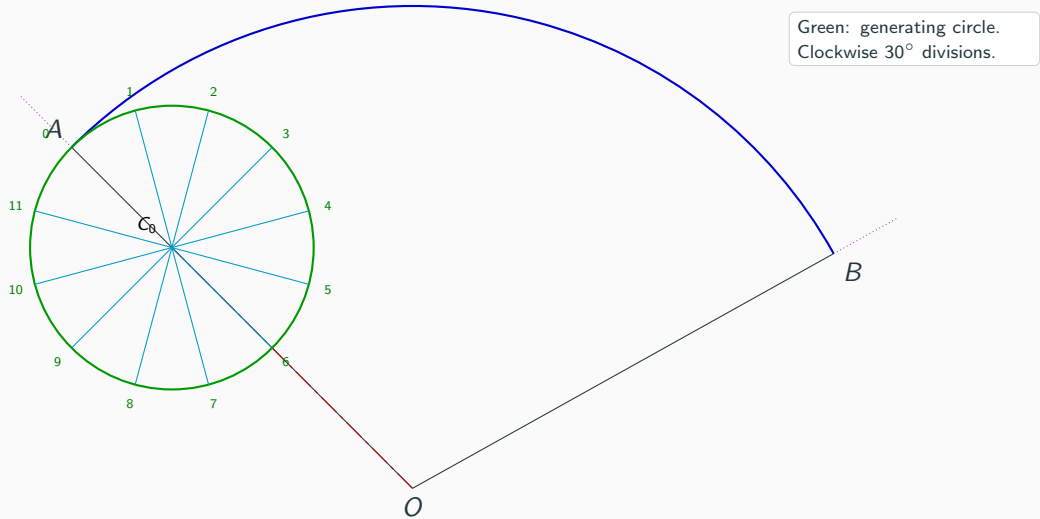
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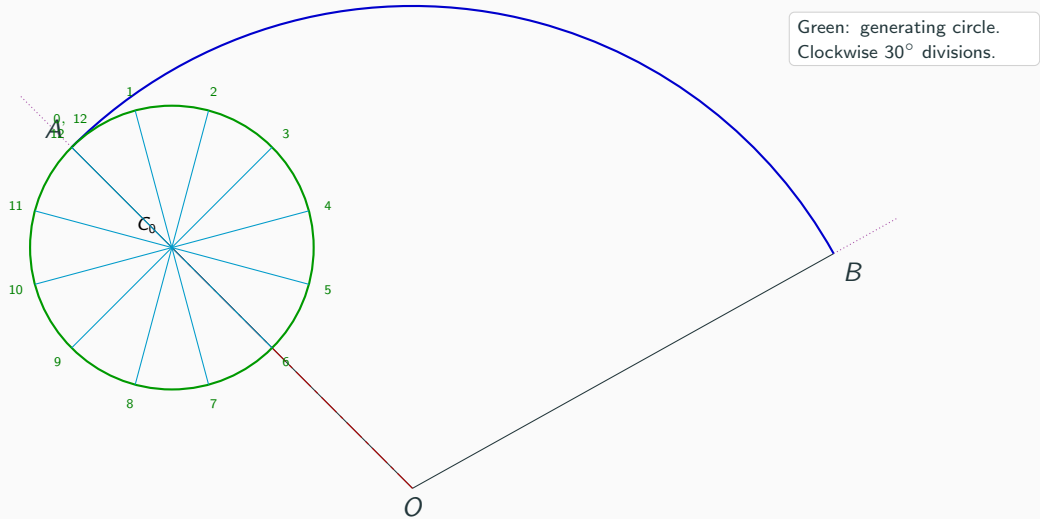
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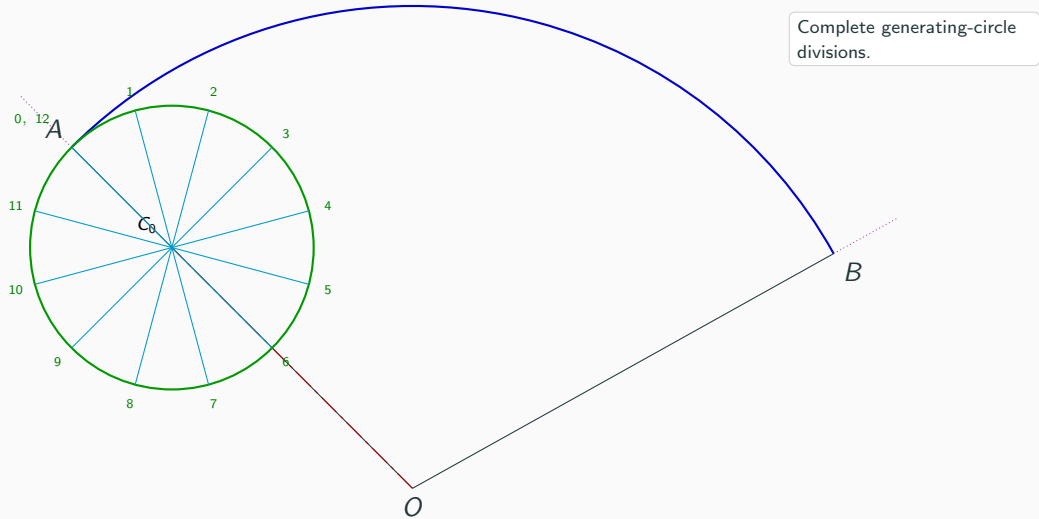
Step 5: Divide the generating circle into 12 equal parts



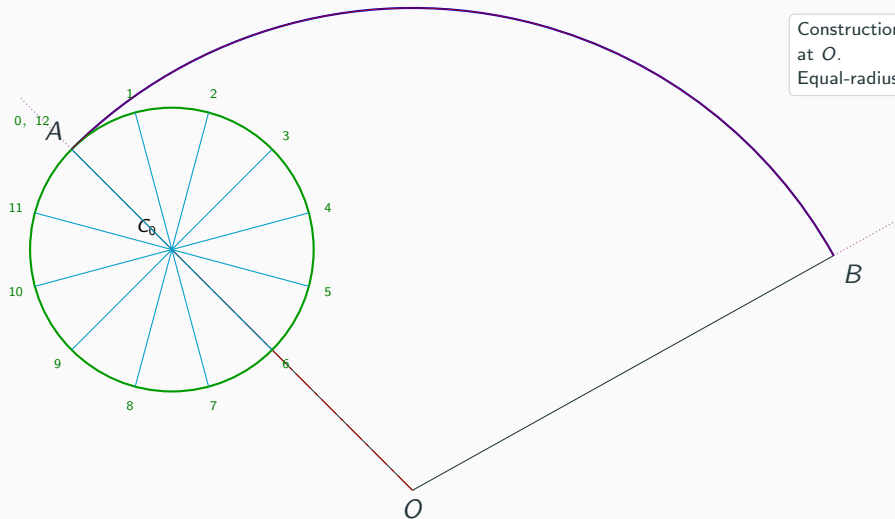
Step 5: Divide the generating circle into 12 equal parts



Step 6: Complete the generating-circle divisions

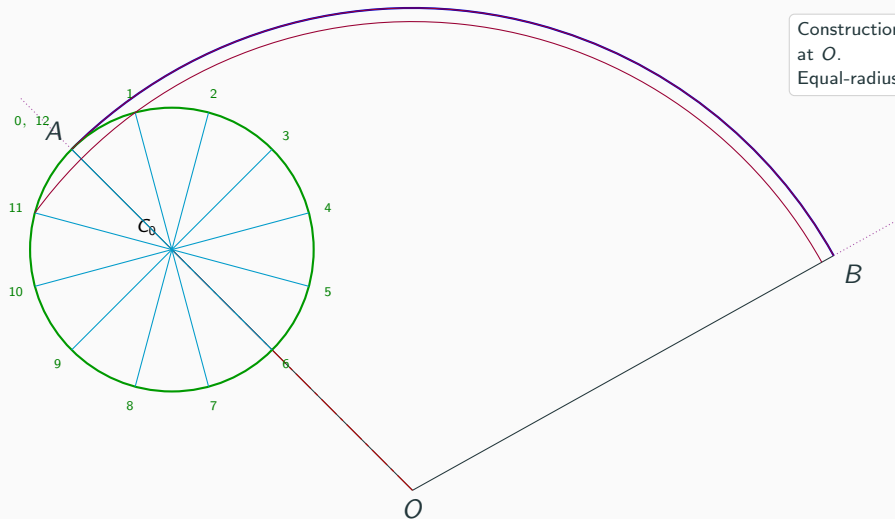


Step 7: Draw concentric construction arcs

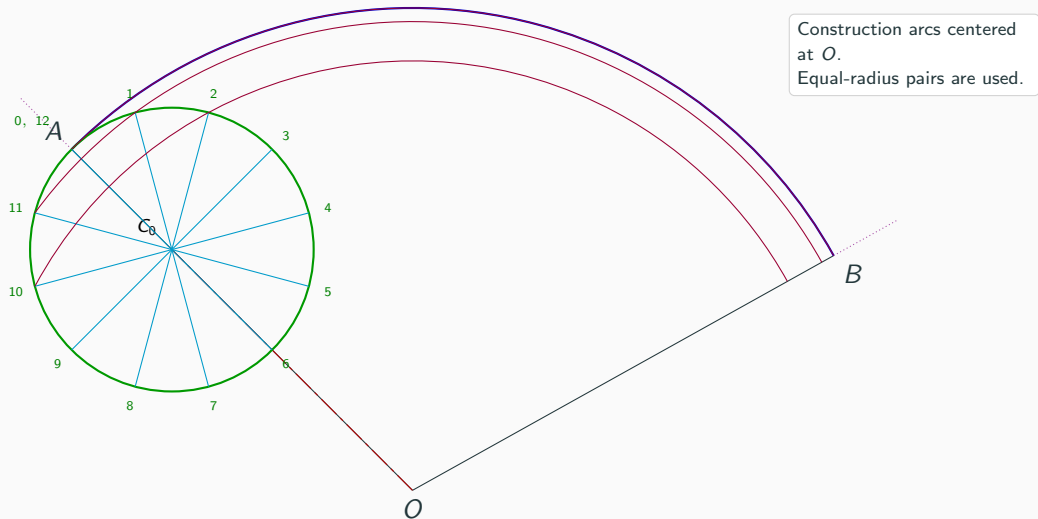


Construction arcs centered at O .
Equal-radius pairs are used.

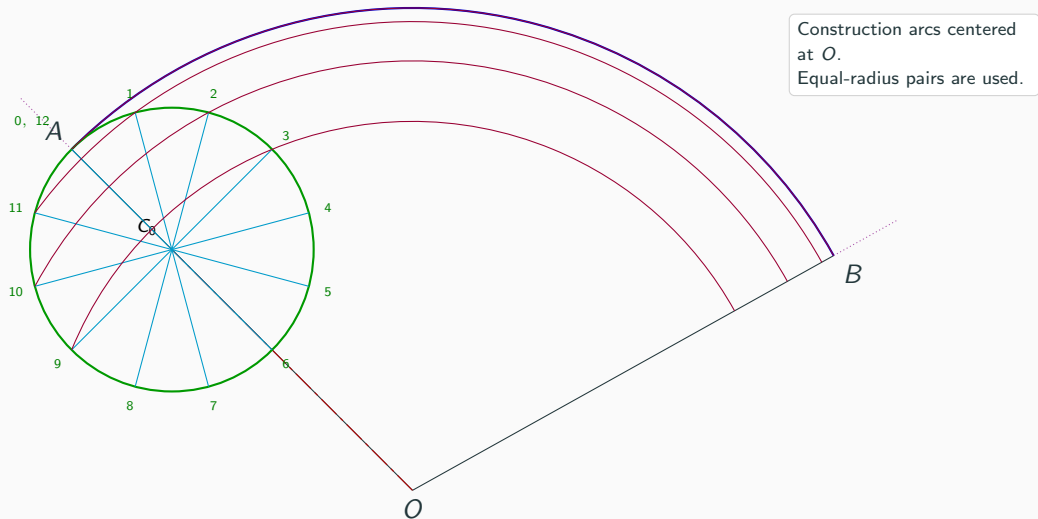
Step 7: Draw concentric construction arcs



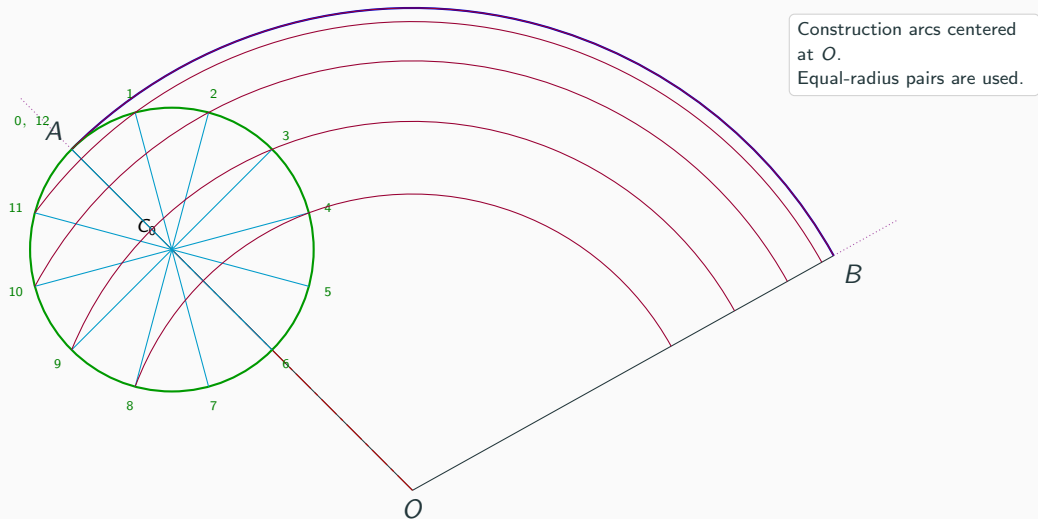
Step 7: Draw concentric construction arcs



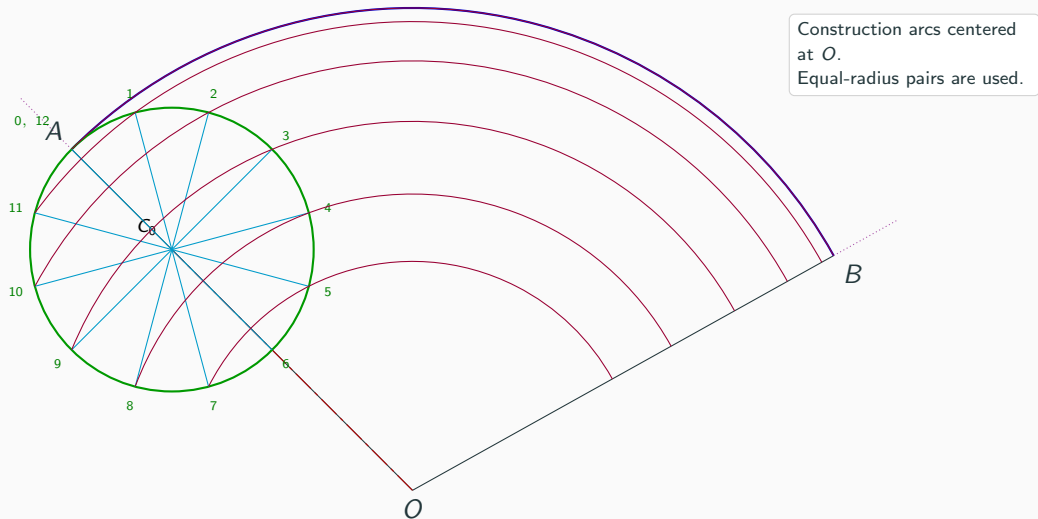
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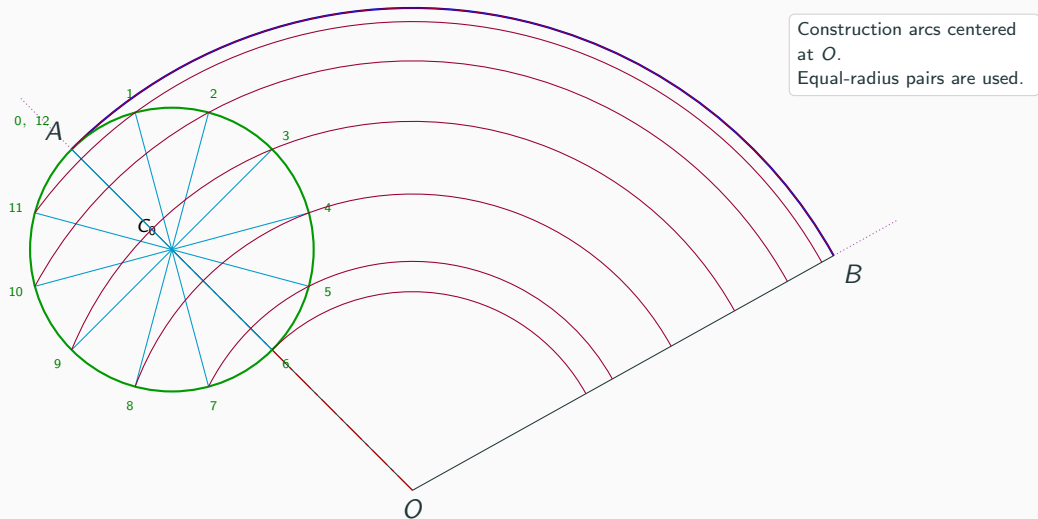
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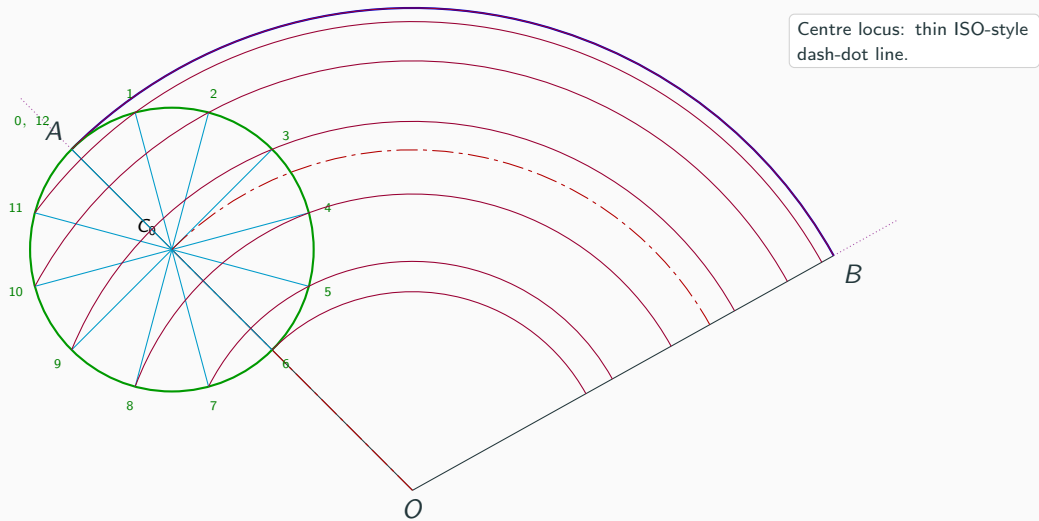
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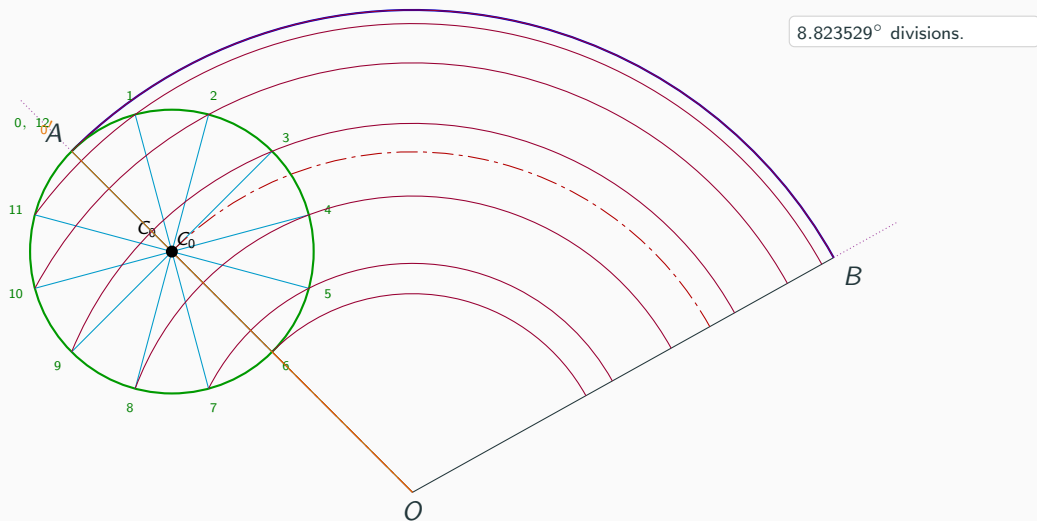
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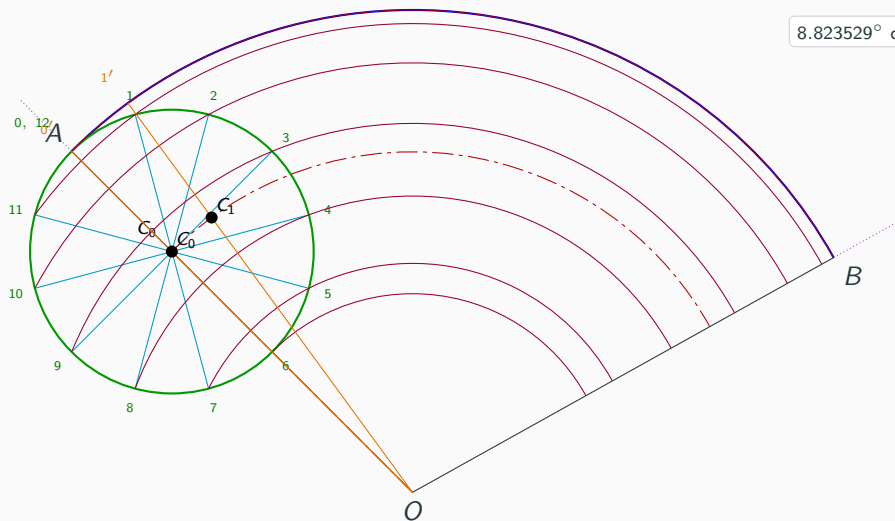
Step 8: Draw the centre arc through C_0



Step 9: Divide $\angle AOB$ into 12 equal parts

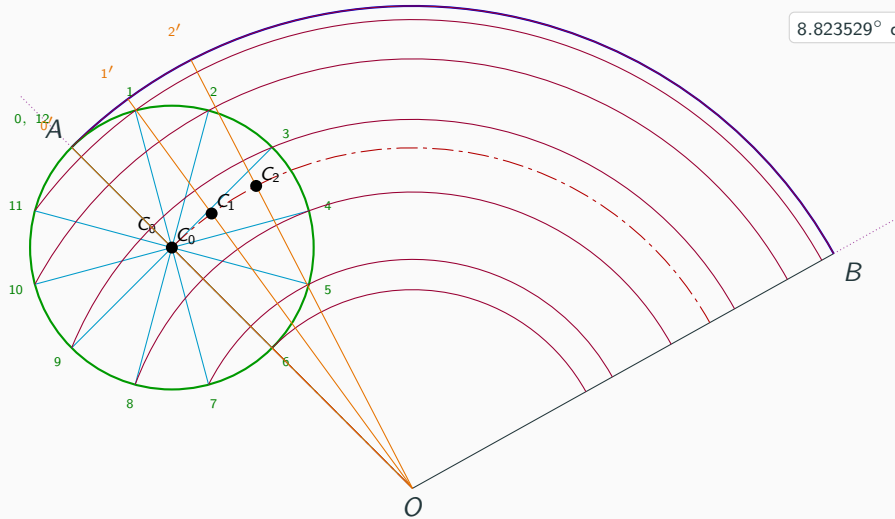


Step 9: Divide $\angle AOB$ into 12 equal parts



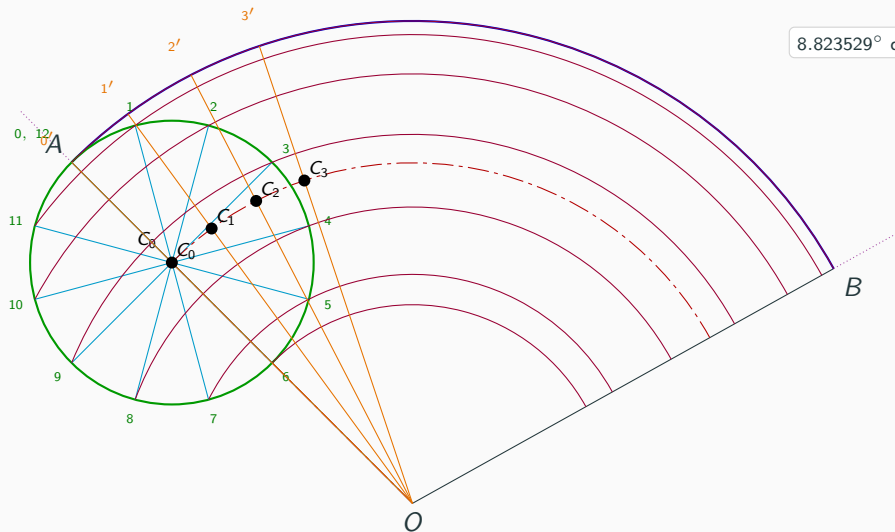
8.823529° divisions.

Step 9: Divide $\angle AOB$ into 12 equal parts



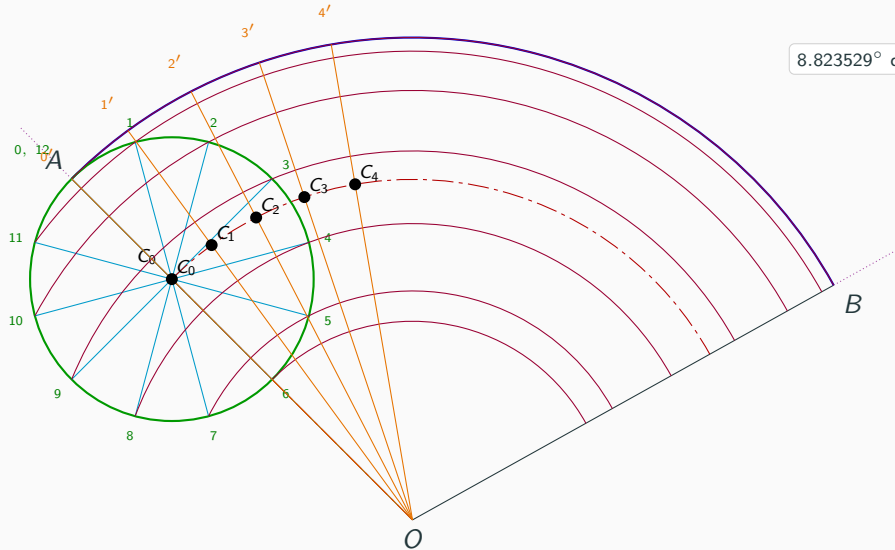
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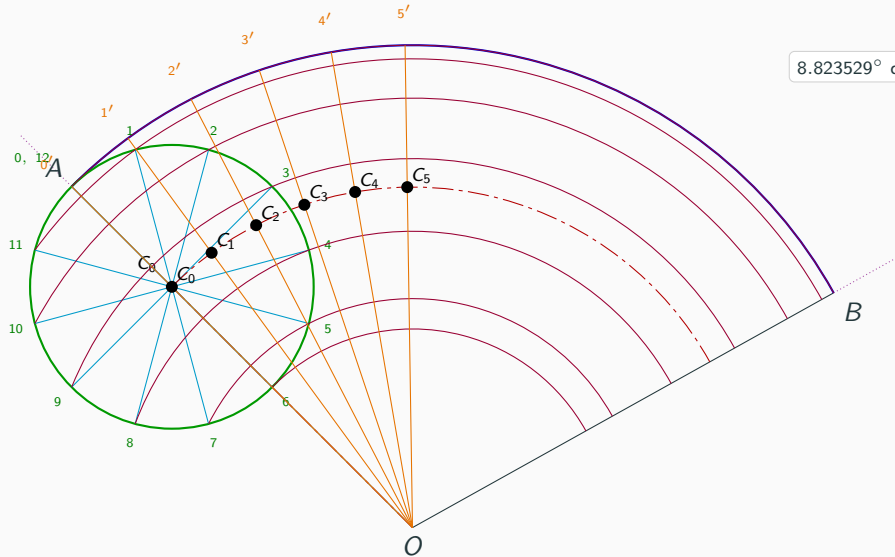
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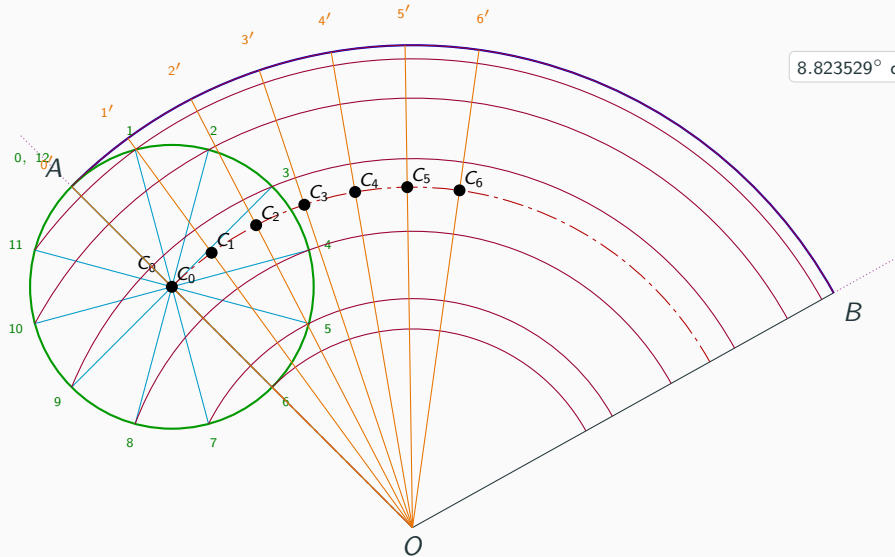


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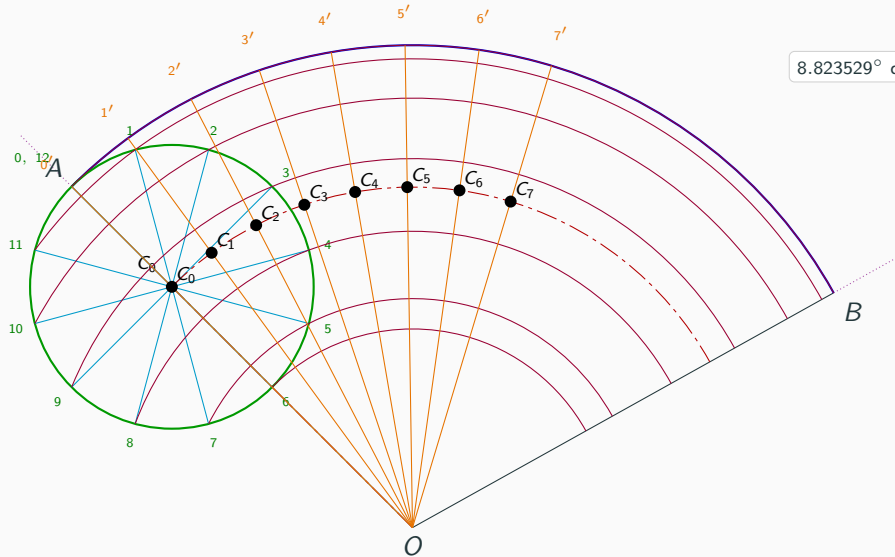
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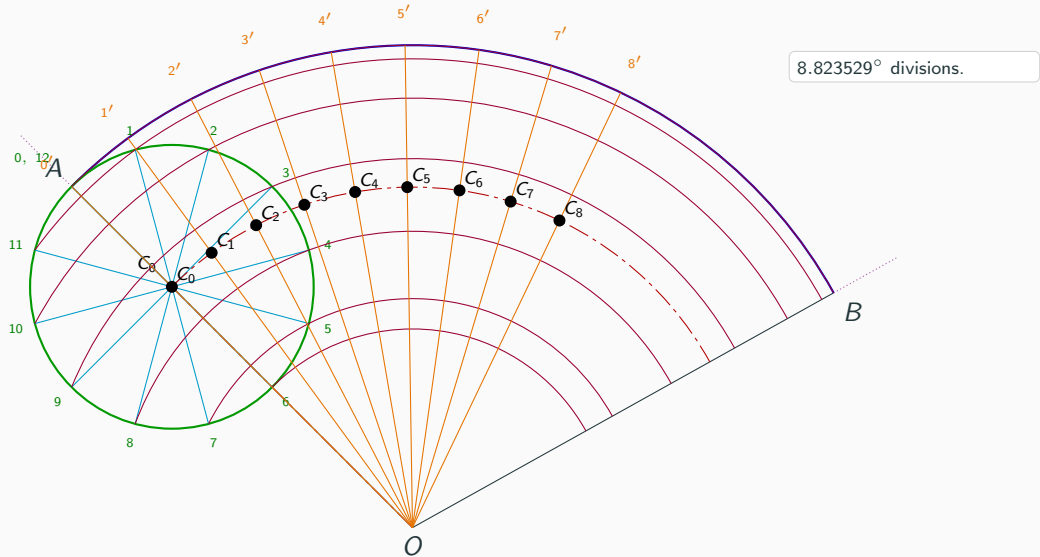


Step 9: Divide $\angle AOB$ into 12 equal parts

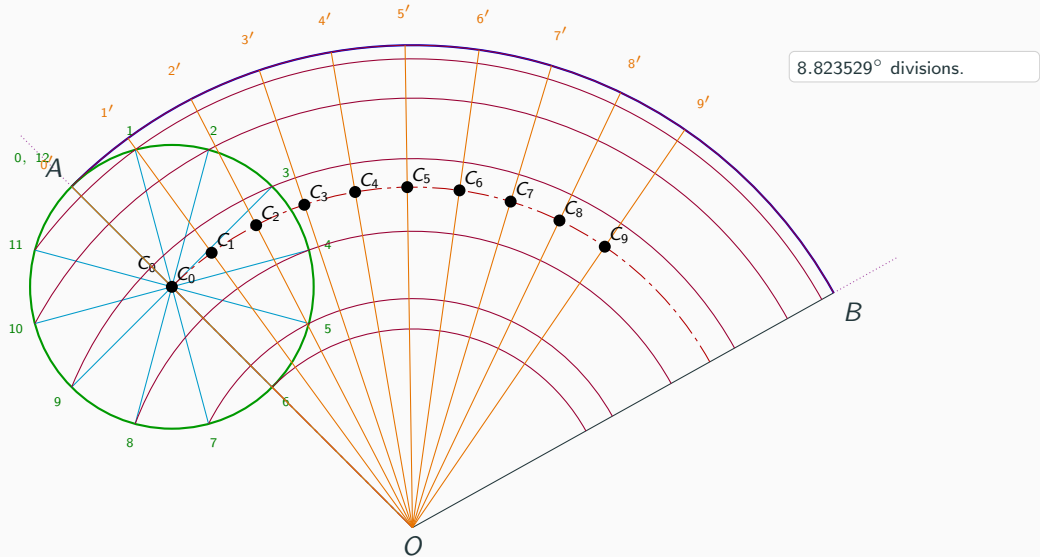


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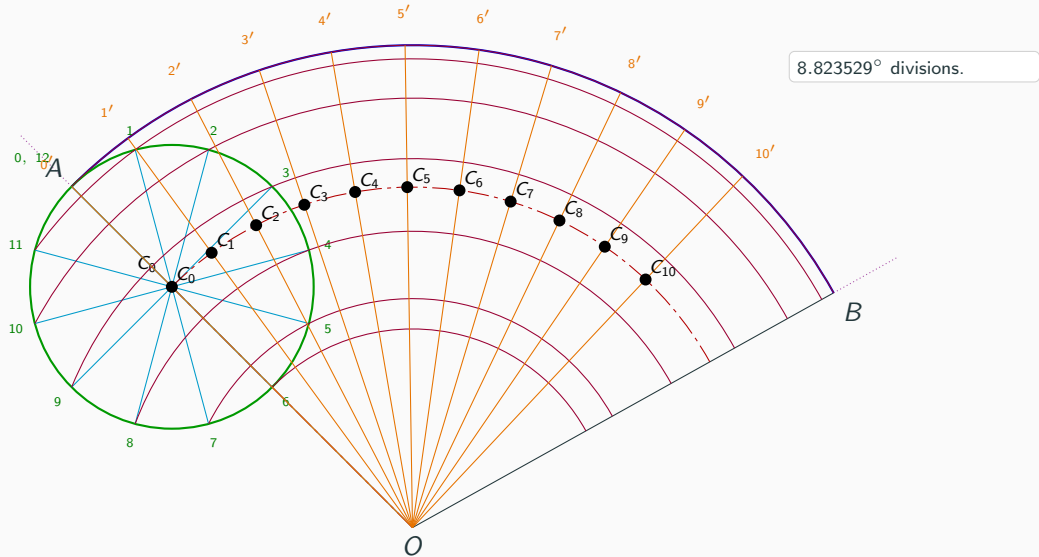
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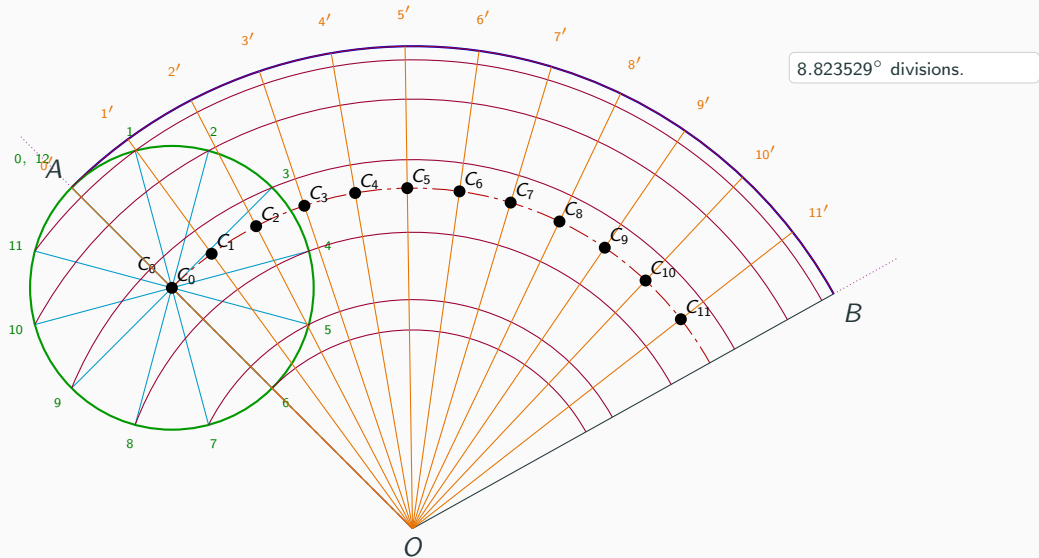
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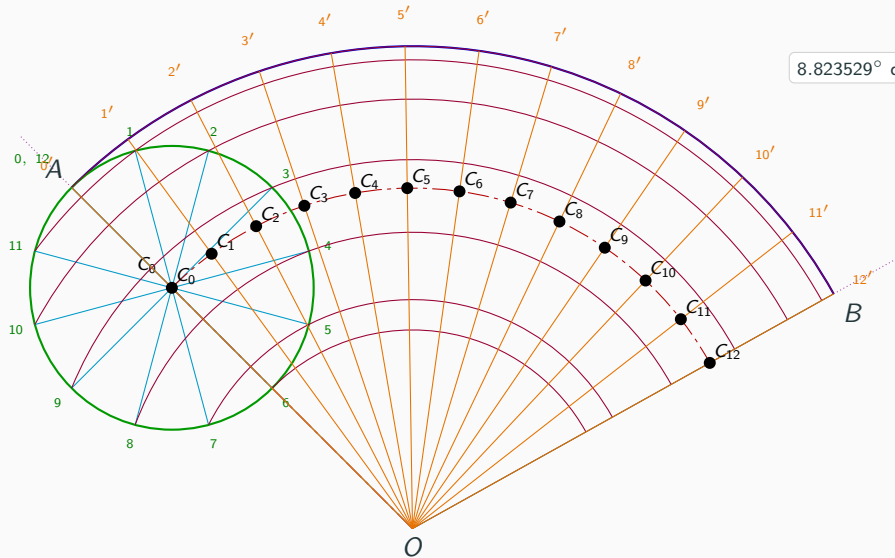
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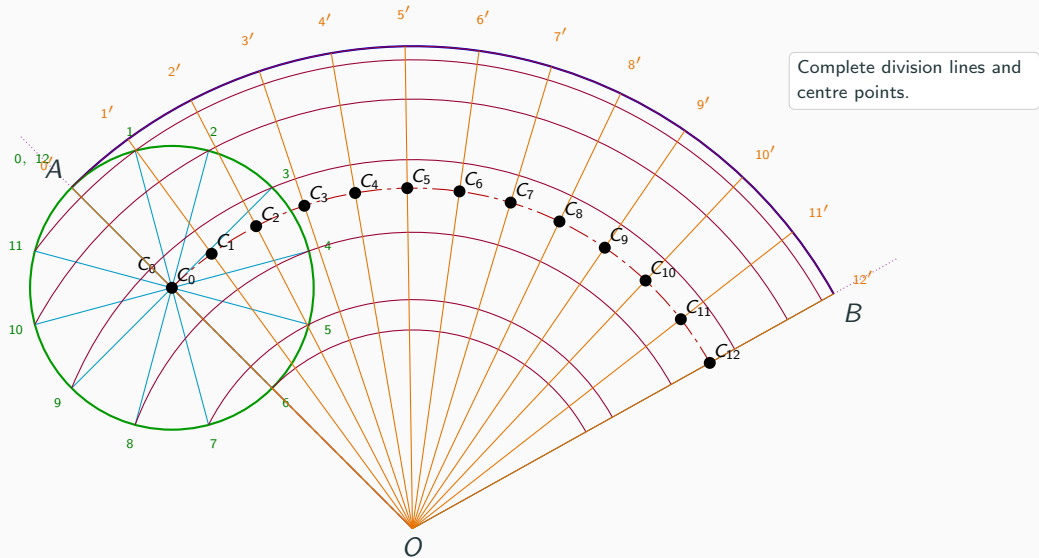
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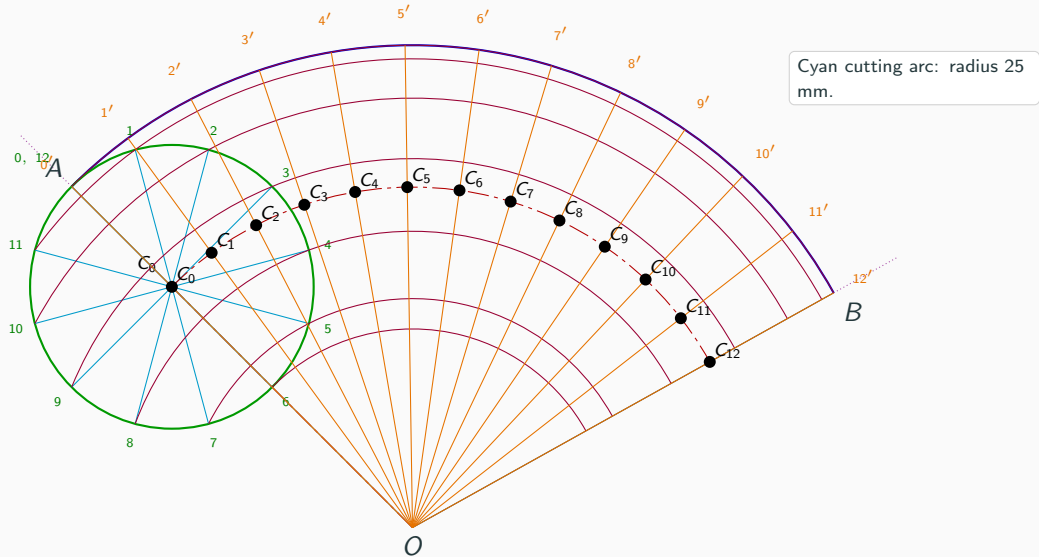
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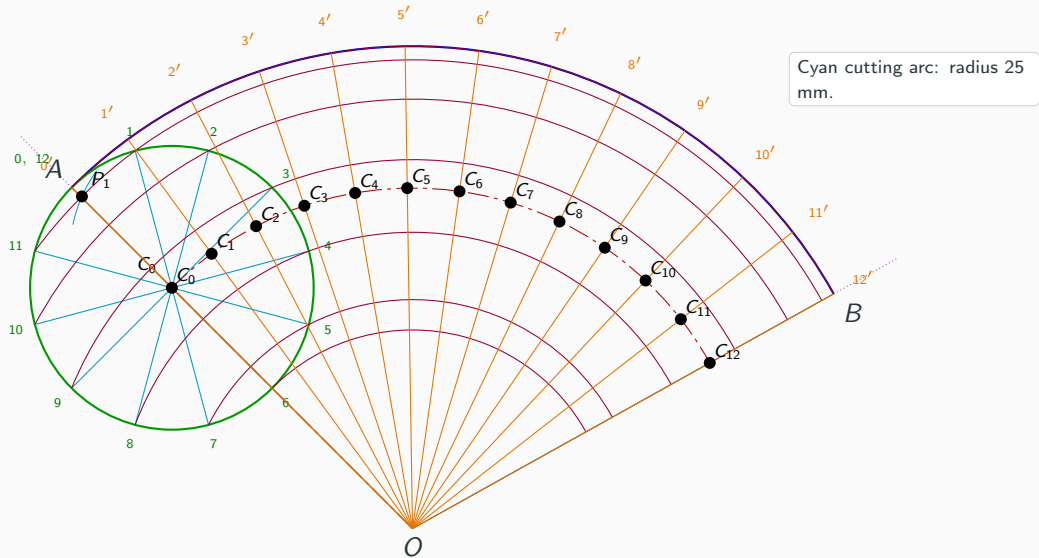
Step 10: Complete the angular divisions and centre points



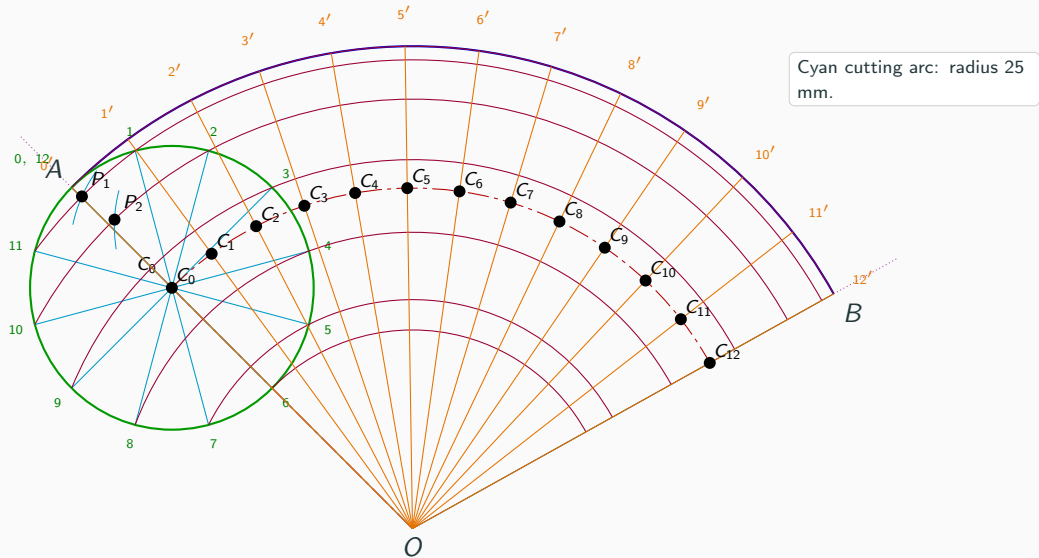
Step 11: Draw the cutting arc and locate P_i



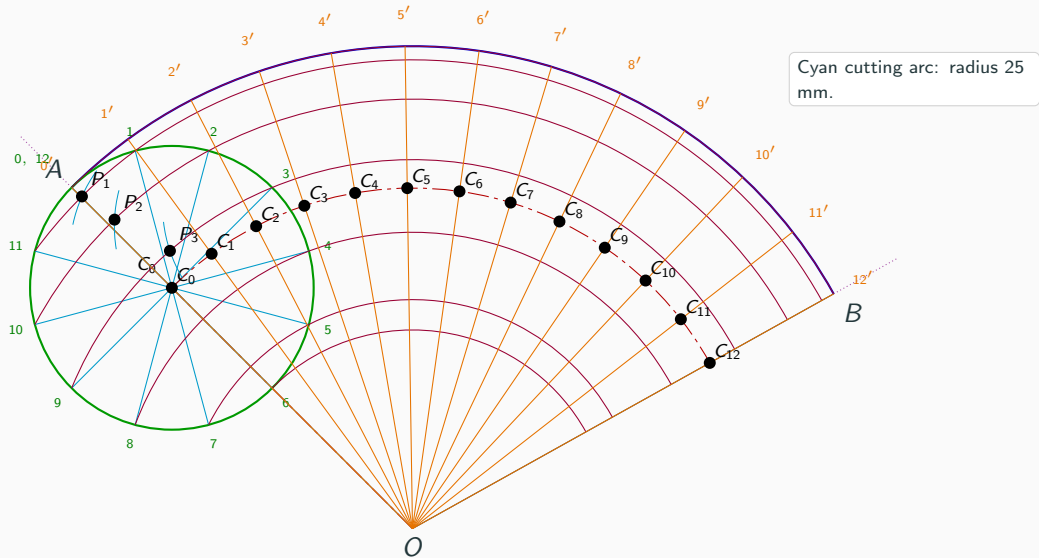
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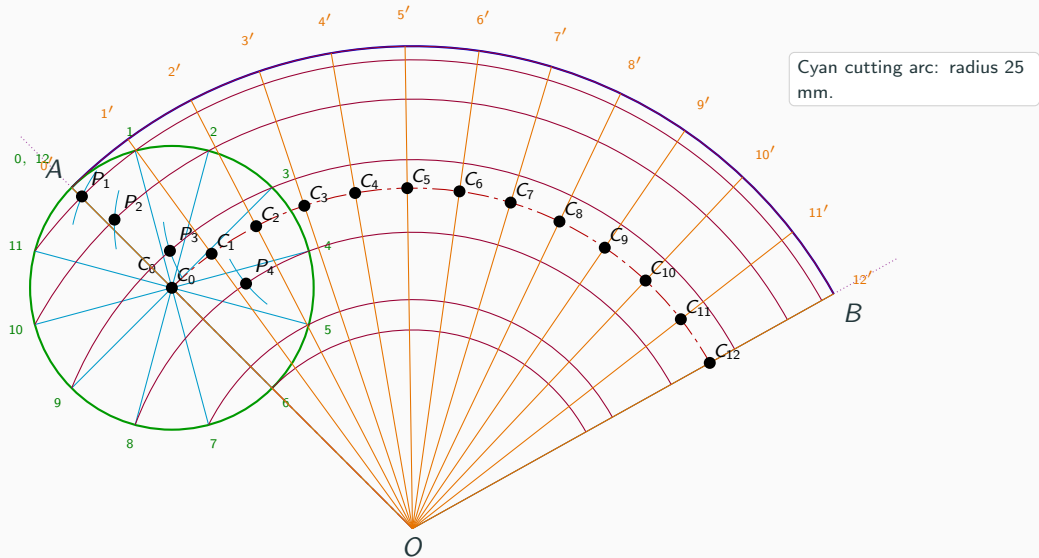
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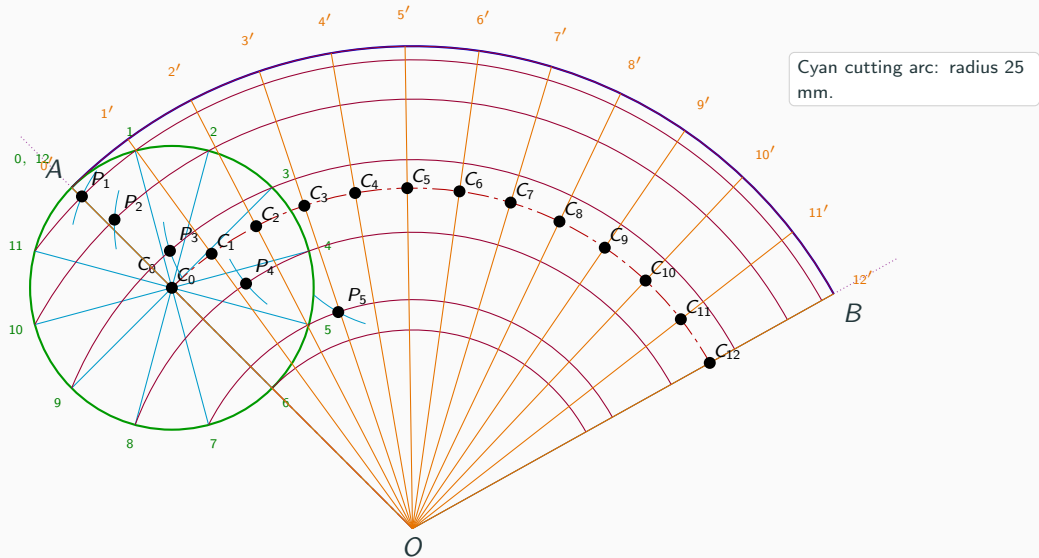
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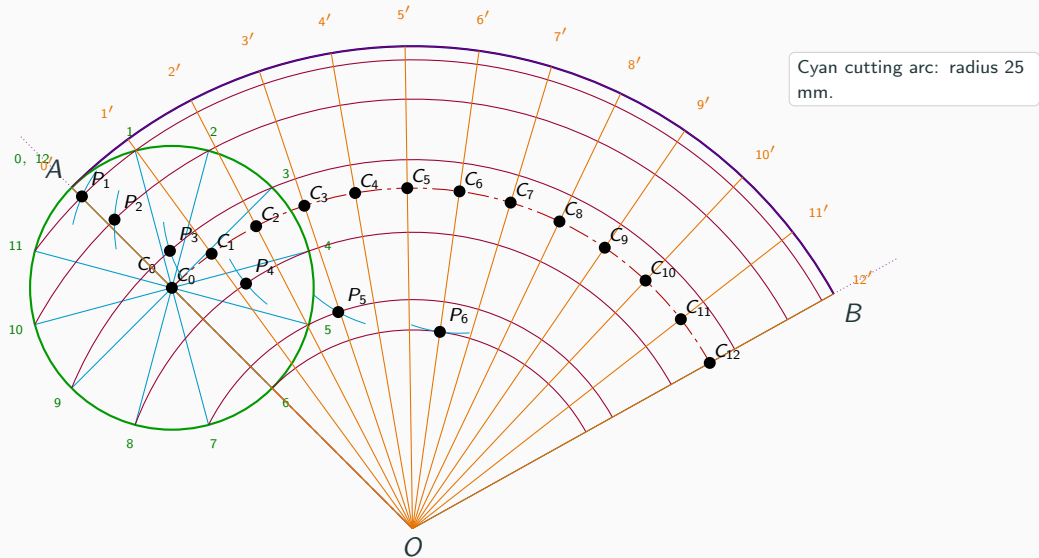
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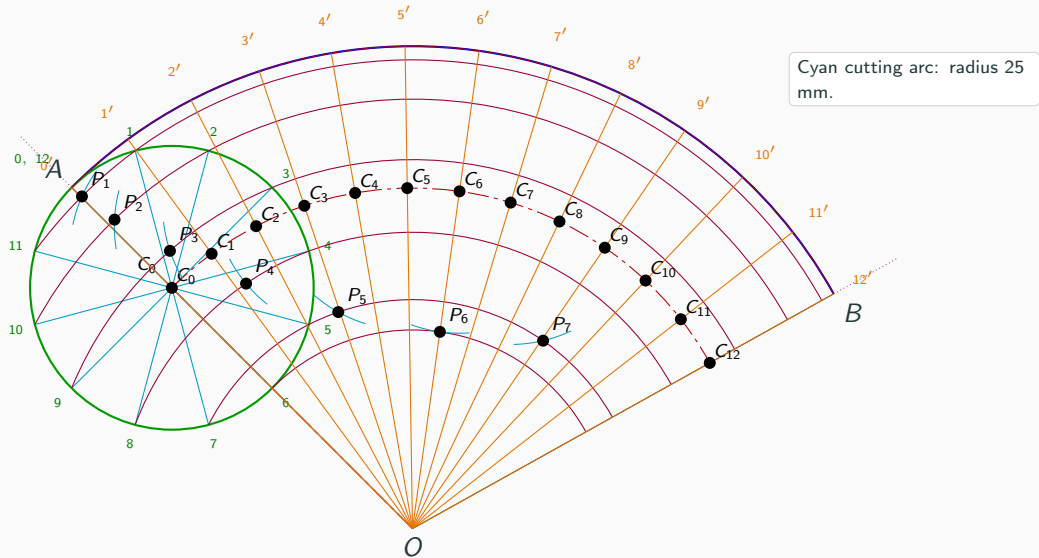
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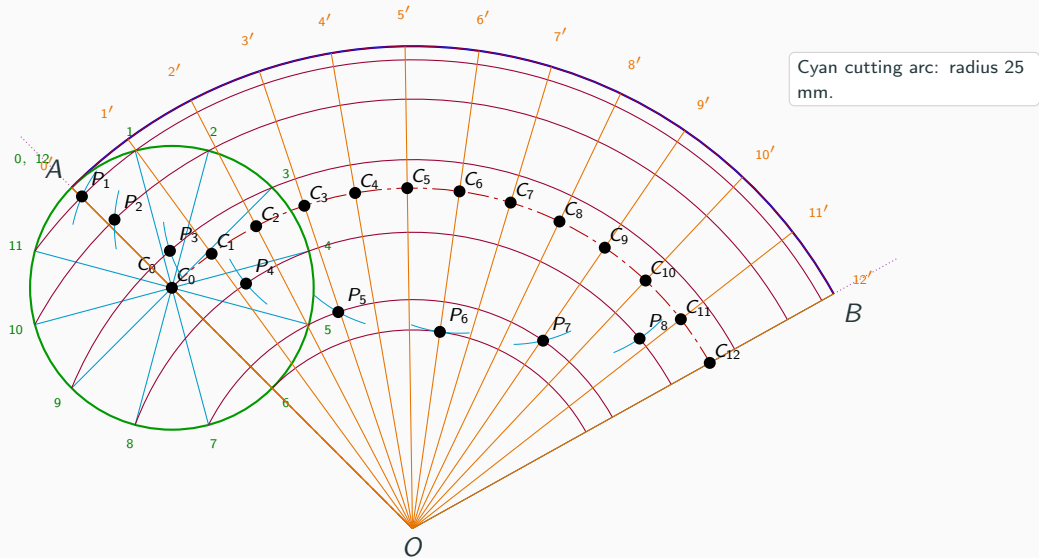
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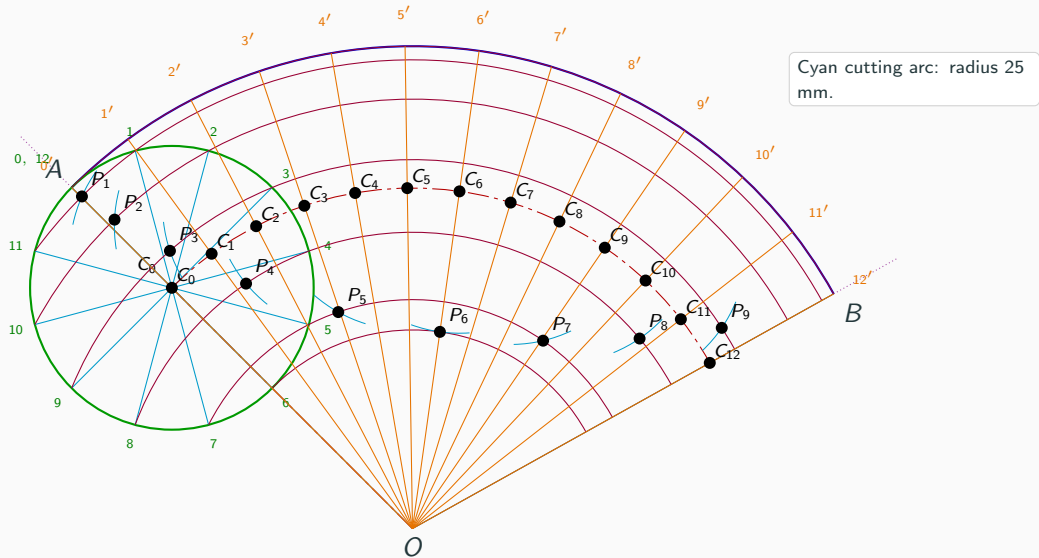
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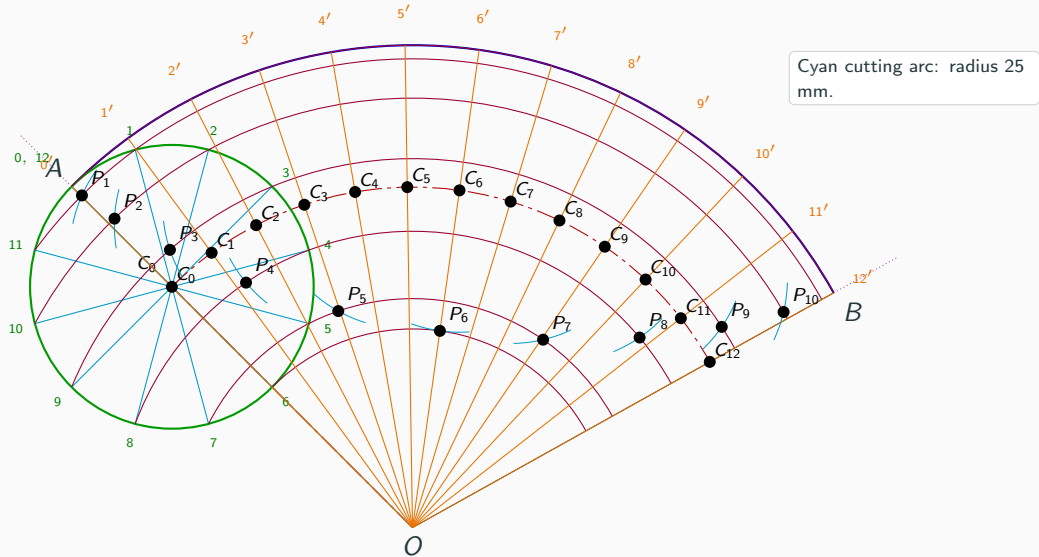
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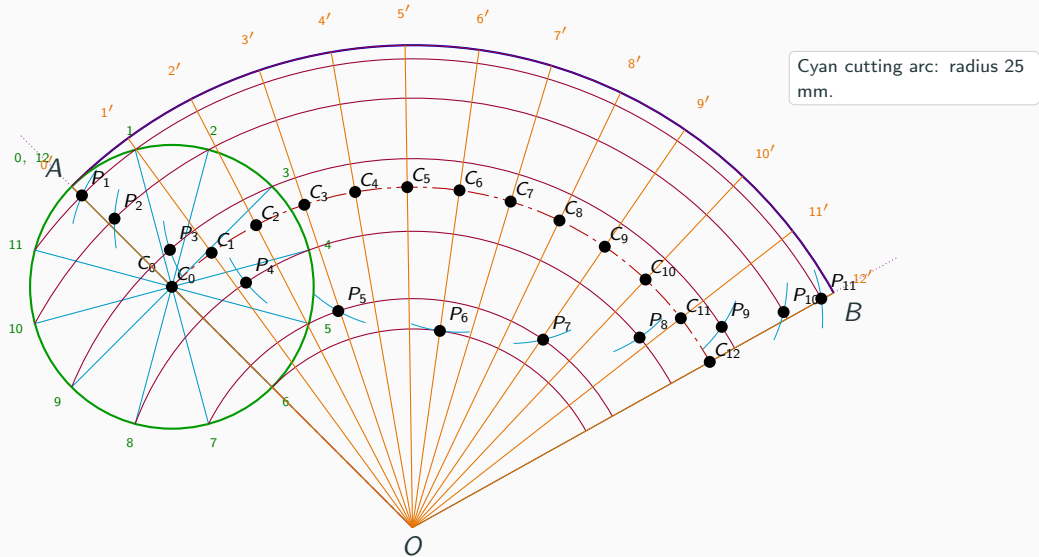
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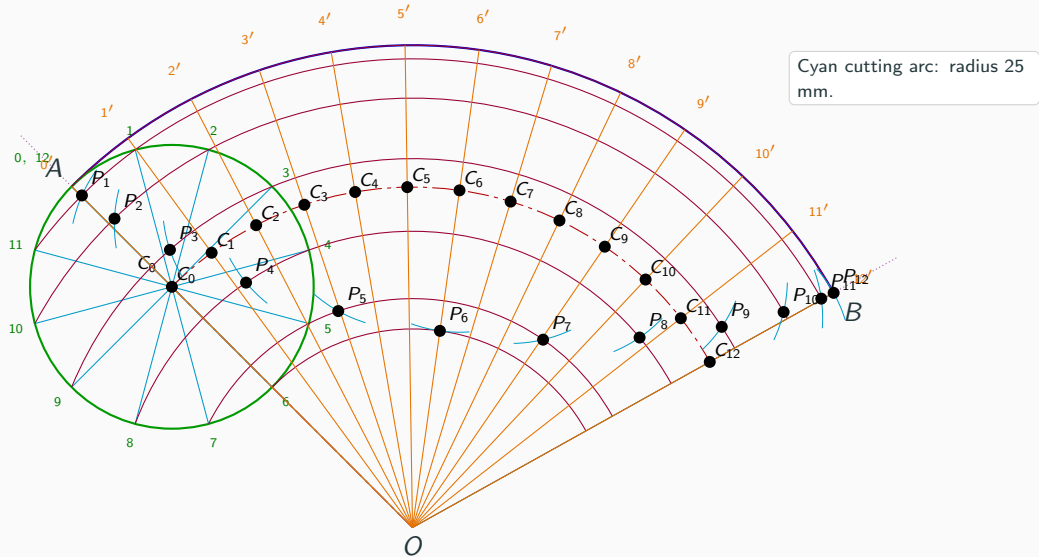
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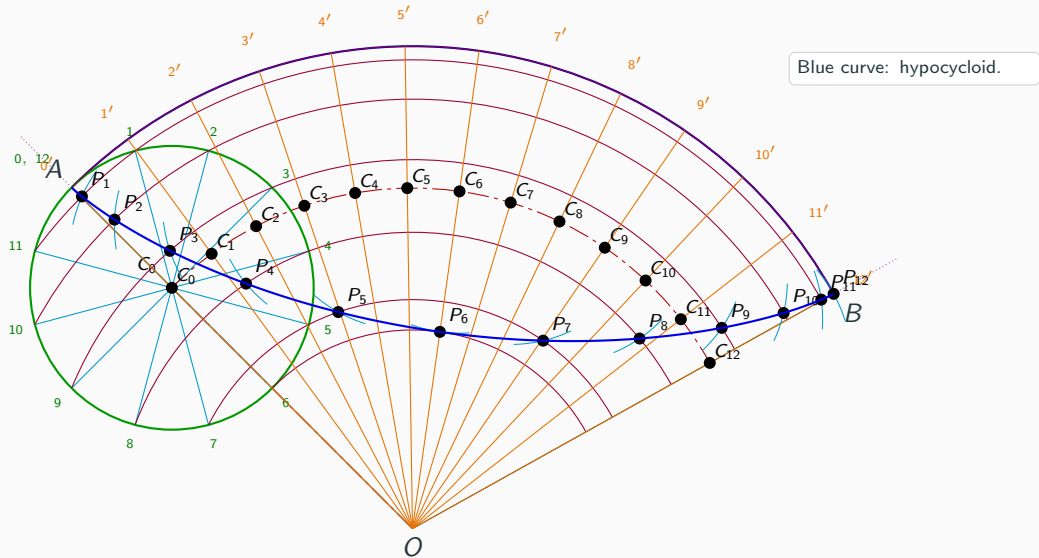
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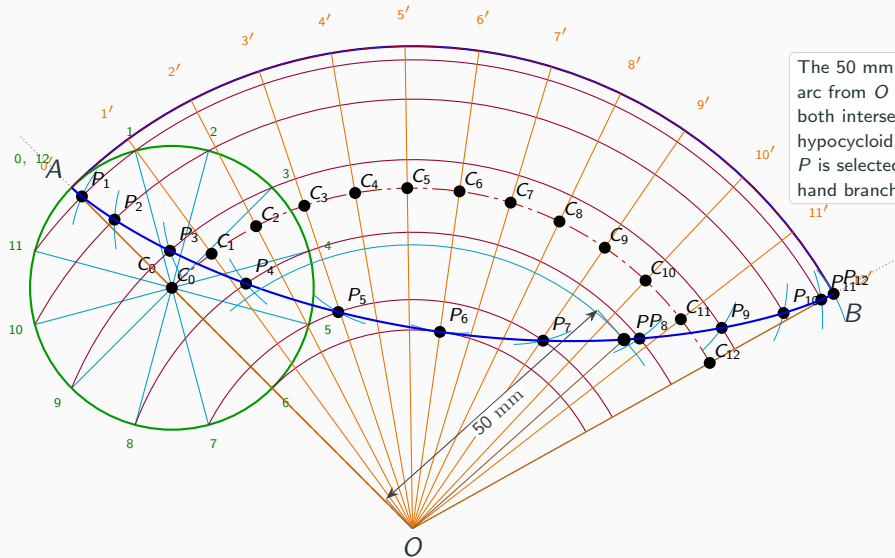
Step 11: Draw the cutting arc and locate P_i



Step 12: Draw the hypocycloid

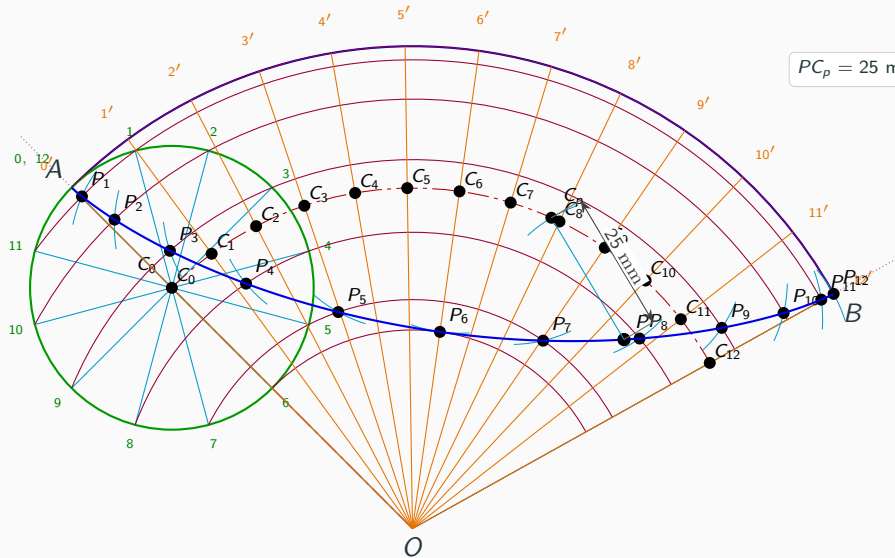


Step 13: Locate P at 50 mm from O

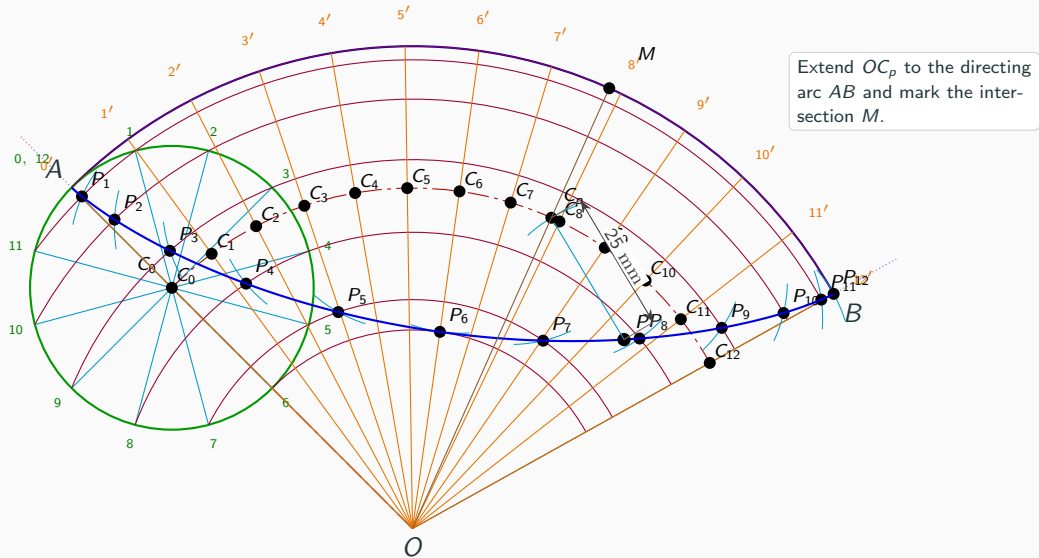


The 50 mm construction arc from O extends beyond both intersections with the hypocycloid. P is selected on the right-hand branch.

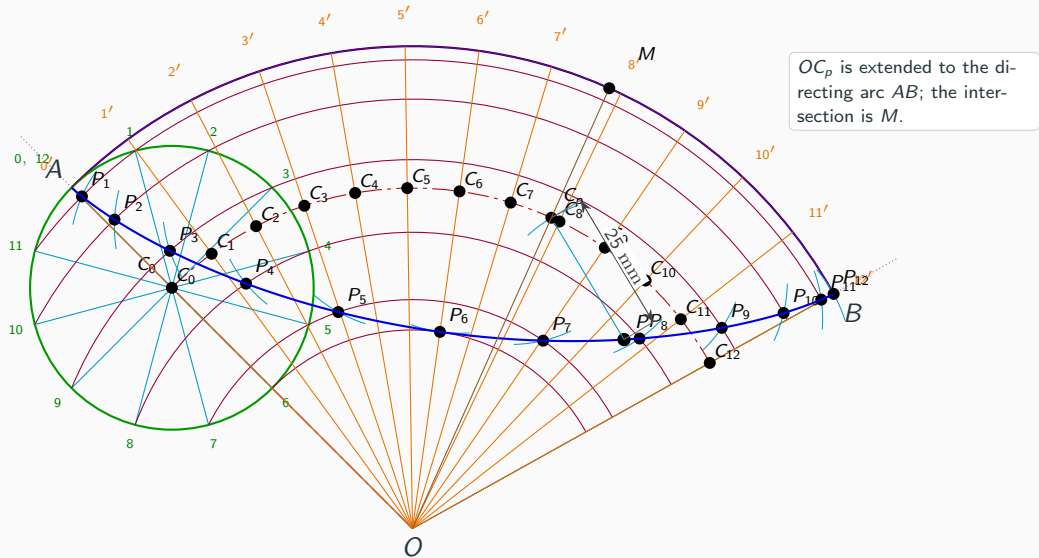
Step 14: Draw the 25 mm radius to C_p



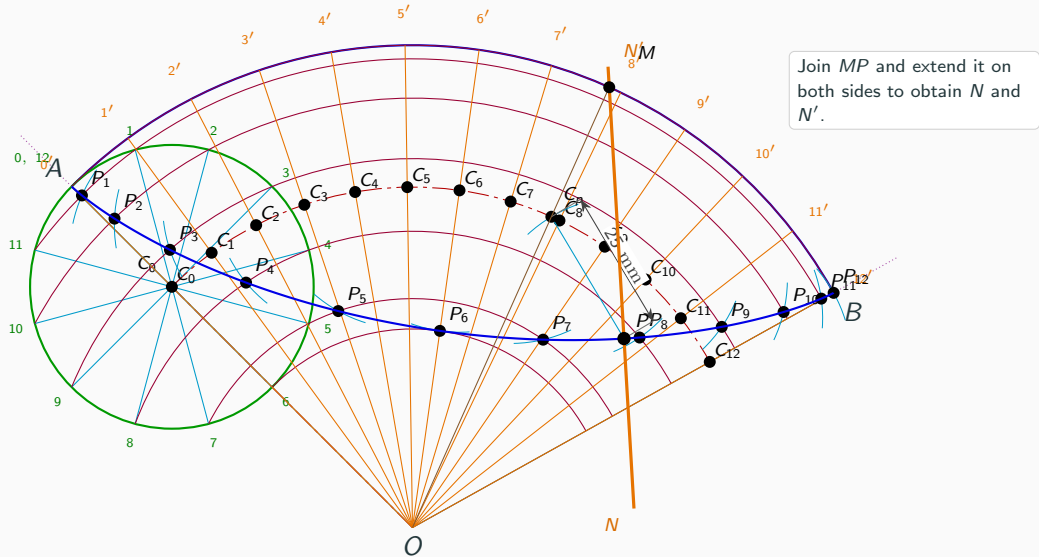
Step 15: Extend OC_p to the directing arc AB and mark the intersection M



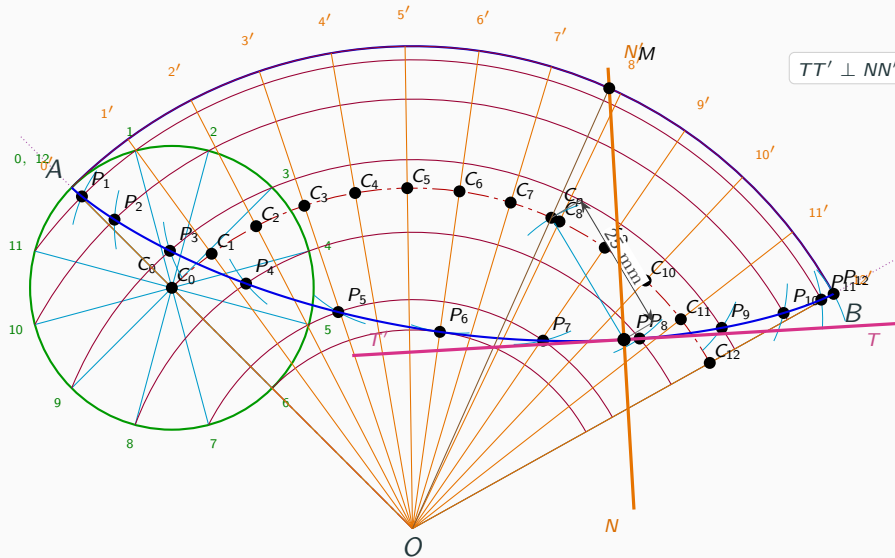
Step 16: Mark M on the directing circle



Step 17: Draw the normal NN' through P

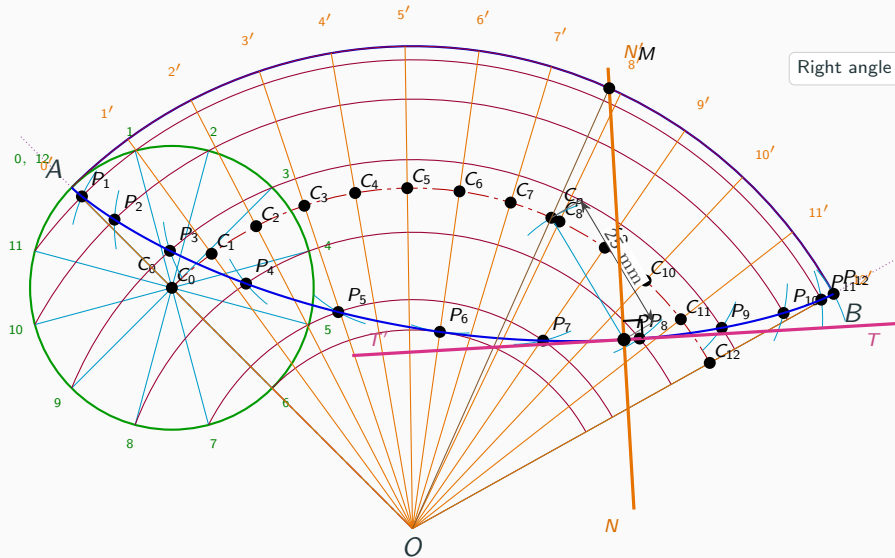


Step 18: Draw the tangent TT' perpendicular to NN'



$TT' \perp NN'$ at P .

Step 19: Show the right angle at P



Right angle at P .

Step 20: Final completed hypocycloid construction

